Sadia Asif

+1 (838) 217-0530 | asifs@rpi.edu | sadiaasif492@gmail.com | Website | Google Scholar | GitHub | LinkedIn

Summary

- **Objective**: Seeking a research based position with a professional and work driven environment where I can utilize and apply my knowledge which would enable me to grow while fulfilling organizational goals.

Education

- **Rensselaer Polytechnic Institute** — Troy, NY, USA
  PhD in Computer Science — *2024 - Present*
  - **Research Focus**: Trustworthy and privacy-preserving AI, focusing on machine unlearning, safeguarding multi-agent LLM architectures, information-theoretic privacy control, RL for agents safety and safety evaluation of large foundation models and agentic systems
- **National University of Sciences and Technology (NUST)** — Islamabad, Pakistan
  B.Sc. Electrical Engineering and Minor in Computer Science — *2017 - 2021*
  - **Thesis**: Developed a deep learning-based EEG abnormality detection system to enhance diagnostic accuracy for healthcare applications.

Publications

- **BLINDSPOT: A Benchmark for Safety and Refusal Calibration in Long-Horizon Tool-Using Agents**: **Sadia Asif**, Mohammad Mohammadi Amiri, Momin Abbas, Tejaswini Pedapati, Prasanna Sattigeri Under Review (ICLR 2027).
- **RISA: Risk-Informed Safety Activation for Long-Horizon Tool Using Agents**: **Sadia Asif**, Mohammad Mohammadi Amiri, Momin Abbas, Tejaswini Pedapati, Prasanna Sattigeri, Karthikeyan Natesan Ramamurthy Under Review (ICLR 2027). arXiv:2605.22786.
- **LCGuard: Latent Communication Guard for Safe KV Sharing in Multi-Agent Systems**: **Sadia Asif**, Mohammad Mohammadi Amiri, Momin Abbas, Prasanna Sattigeri, Karthikeyan Natesan Ramamurthy Under Review (Neurips 2026). arXiv:2605.22786.
- **Information-Theoretic Privacy Control for Sequential Multi-Agent LLM systems**: **Sadia Asif**, Mohammad Mohammadi Amiri. Under Review (Neurips 2026). arXiv:2603.05520.
- **OFMU: Optimization-Driven Framework for Machine Unlearning**: **Sadia Asif**, Mohammad Mohammadi Amiri. ICLR 2026. openreview.net/forum?id=ZDuyNJI56H.
- **RefusalGuard: Geometry-Preserving Fine-Tuning for Safety in LLMs**: **Sadia Asif**, Mohammad Mohammadi Amiri. COLM 2026. arXiv:2605.01913.
- **When LLMs Manipulate: Measuring Behavioral, Social, and Temporal Effects**: **Sadia Asif**, Israel Antonio Rosales Laguan, Shumaila Asif, Haris Khan, Muneeb Asif, Muhammad Zeeshan Karamat. NeurIPS 2026 Workshop on Dynamic Alignment in Human–AI Coupled Systems, (2026).
- **Generative AI Agents for Controllable and Protected Content Creation**: Haris Khan, **Sadia Asif**. NeurIPS 2025 Workshop on Generative AI for Protected and Controllable Content (GenProCC), 2025.
- **Privacy, Security, and Data Leakage Concerns in Multi-Agent AI Systems: A Literature Review**: Muhammad Zeeshan Karamat, **Sadia Asif**, Haris Khan, Muhammad Shaheer, Shumaila Asif, Rajesh Upadhayaya, Christiana Chamon, Mohammad Mohammadi Amiri. Under Review.
- **DarkPatterns-LLM: A Comprehensive Benchmark for Detecting Manipulative and Harmful Behaviors in Large Language Models**: **Sadia Asif**, Israel Loguan, Shumaila Asif, Haris Khan. Under review.
- **Modular Delta Merging with Orthogonal Constraints: A Scalable Framework for Continual and Reversible Model Composition**: Haris Khan, **Sadia Asif**, Shumaila Asif. NewinML @ Neurips 2026 .
- **Multi-Agent Framework for Controllable and Protected Generative Content Creation: Addressing Copyright and Provenance in AI-Generated Media**: Haris Khan, **Sadia Asif**. **IEEE International Conference on Data Mining Workshops (ICDMW 2025)**. ieee.org/abstract/document/11416432.

Experience

Research Extern, IBM Research — Yorktown Heights, USA
IBM Mentors: Prasanna Sattigeri, Karthikeyan Natesan Ramamurthy, Momin Abbas — *May 2026 – Aug 2026*

- **Long-Horizon Multi-Agent Risk Trajectory Prediction**: Developing methods for predicting emerging safety risks and unsafe trajectories across multi-turn interactions in multi-agent tool-calling systems.
- **Long-Horizon Agent Safety Benchmark**: Developing a trajectory-level benchmark for evaluating safety in long-horizon tool-using agents across safe completion, unsafe completion, correct refusal, over-refusal, and indeterminate outcomes under diverse multi-turn attack scenarios.

Research Collaborator, IBM Research — Yorktown Heights, USA
IBM Mentors: Prasanna Sattigeri, Karthikeyan Natesan Ramamurthy, Momin Abbas — *Jan 2026 – Present*

- **Secure KV Cache Sharing in Multi-Agent LLM Systems**: Investigating security and privacy risks arising from Key–Value (KV) cache sharing in multi-agent and long-context LLM pipelines, including cross-agent information leakage, representation-level inference attacks, and side-channel vulnerabilities.
- **Threat Modeling and Empirical Evaluation**: Designing formal threat models and conducting empirical studies to quantify leakage from shared internal representations, with the goal of developing leakage-bounded and privacy-aware KV-sharing mechanisms.

Graduate Research Assistant Amiri Lab @ RPI
Advisor: Dr. Mohammad Mohammadi Amiri *Aug 2024 – Present*

- **Machine Unlearning**: Designed theoretically grounded and computationally efficient unlearning algorithms for large language models using penalty-based bi-level optimization to selectively remove targeted knowledge while preserving downstream performance. Demonstrated strong forget-set removal with minimal retain-set degradation, resulting in an ICLR 2026 publication.
- **Privacy-Preserving Multi-Agent LLM Systems**: Developed an information-theoretic framework for privacy control in sequential multi-agent LLM pipelines, formally analyzing representation leakage and safety–utility trade-offs. Validated effectiveness through empirical leakage estimation and adversarial evaluation. Work under review at ICML 2026.
- **Safe Fine-Tuning of LLMs**: Investigating alignment-preserving fine-tuning strategies based on representation-level constraints to prevent collapse of refusal behaviors and safety boundaries in pretrained foundation models, enabling capability improvement without safety regression.
- **Data Poisoning and Unlearnable Text**: Studying text-based data poisoning and unlearnable example generation methods to detect and discourage unauthorized model training on copyrighted or sensitive datasets, analyzing robustness under adaptive adversaries.
- **Differential Privacy and Data Valuation**: Designed privacy-aware data valuation mechanisms for multi-seller settings using differential privacy, balancing economic incentives with provable privacy guarantees and controlled performance loss.
- **Strategic Planning in LLMs**: Proposed an artificial oracle framework for anticipatory reasoning in multi-turn interactions, modeling user-response prediction via reinforcement learning and Markov decision processes to improve deliberative planning performance.

Head of RnD and Technical Lead Quant Data Solutions
Aug 2021 - Aug 2024

- **Compensation Data Analysis**: Optimized the compensation analysis system for a leading silicon valley based firm by fine-tuning and deploying large language models (LLMs) to enhance efficiency, scalability, and analytical accuracy. Developed advanced tools leveraging LLMs for improved compensation data interpretation and decision-making. Identified and resolved performance bottlenecks, significantly improving system responsiveness and operational efficiency. Designed and implemented an AI-driven agent powered by fine-tuned LLMs to automate compensation data analysis, enabling large-scale processing of employee records and extracting meaningful insights for enhanced predictive modeling.
- **Quantitative Modeling**: Analyzed Historical time series data of different worldwide exchanges to find useful patterns by working with leading HFT firm in Chicago. Implemented novel algorithmic strategies every week based on observed trends in data and after reading various research papers. Applied various statistical and advanced mathematics techniques to find anomalies and to convert observed patterns into mathematical equations and applying Feature Engineering techniques to get hand crafted features for Machine Learning models for price forecasting and developed Quants to be used in live market data for algorithmic trading.
- **AI System Development**: Worked on development of Theft detection end to end system using various state of the art deep learning based models for streaming video surveillance such as Yolo, I3D , assemblnet and MoViNet for building custom model related to ATM crime detection and for other outdoor theft related activities detection.
- **Speaker Identification based AI Module**: Worked on development of novel AI based system which can recognize multiple speakers in noisy audio data, achieving performance that surpassed existing approaches.
- **Backend Development**: Designed and implemented scalable backend systems along with efficient database schema design from scratch using Node.js, TypeScript, Redis, PostgreSQL and third party integrations.
- **Tech Lead**: Led technical teams across diverse tech stacks, providing guidance in code review, bug fixing, and logic building to enhance system efficiency and maintain high-quality standards. Focused on performance optimization by identifying inefficiencies, refining system logic, and implementing best practices to improve scalability and reliability.

Founding Engineer E-Mareez Care
Feb 2022 - Present

- **Backend Architecture (Python, FastAPI, Flask, Node.js, Express.js, TypeScript)**: Designed and implemented a scalable, high-performance backend architecture for healthcare applications, ensuring optimal efficiency.
- **System Optimization (PostgreSQL, Redis)**: Engineered seamless PostgreSQL and Redis integrations to enable fast, reliable, real-time data processing for enhanced data integrity and speed.
- **Monitoring Logging (Production Environment)**: Integrated robust monitoring and logging solutions to proactively manage system health, reduce incident response time, and support uninterrupted live production services.
- **Frontend Architecture (React.js, Next.js, TypeScript)**: Worked with other junior developers and designers to ensure seamless integration and development of plateform.

Research Fellow Kyung Hee University AI
Robotics Research Lab · Part-time
Jun 2021 - Sep 2021

- **EEG Motor Imagery for Brain-Controlled Drones**: Developed deep learning algorithms for brain-controlled drones using EEG motor imagery data, advancing human-machine interaction research.
- **Adversarial Attack Research**: Engineered adversarial attack techniques on EEG datasets, evaluating the robustness of models such as EEGNet, VGG, Deep CNN, Shallow CNN, Tidenet, and ResNet.
- **Dataset Utilization**: Worked with EEG datasets, including Physionet, Amigos, and Deap, to design and validate models under real-world scenarios.

Research Assistant National University of Sciences and Technology (NUST)
SIGMA Research Lab *Aug 2020 - Dec 2021*

- **Deep Learning for EEG Analysis**: Localized a deep learning-based algorithm for detecting EEG signal abnormalities and implemented gaze-tracking techniques for EEG signal analysis.
- **Visualization Tools**: Developed a customized tool for visualizing and annotating EEG signal abnormalities to aid in diagnostic workflows.
- **Mentorship**: Supervised five interns on research projects, guiding them in implementing algorithms and preparing evaluations for project milestones.
- **Research Meetings**: Organized and conducted regular progress meetings to evaluate performance and refine research strategies.

Technical Skills

- **Programming**: Python, C/C++, JavaScript, TypeScript, SQL, .Net, Matlab, JAX
- **Frameworks/Libraries**: TensorFlow, PyTorch, HuggingFace, vLLM, Scikit-learn, LlamaIndex, SmolAgents, Langchain, Pandas, Numpy, Scikir-learn, OpenCV,SciPy, RAG, StreamLit, CrewAI, Node.js, Nextjs, React, React Native, Flask
- **Large Language Models**: LLaMA 3.1, Qwen 2.5, Mistral, LongChat, DeepSeek-R1 (inference); long-context and multi-agent LLM pipelines
- **Fine-Tuning & Adaptation**: LoRA, Prefix/Prompt Tuning, Soft Prompting, Representation-Constrained Fine-Tuning
- **Privacy & Alignment**: Differential Privacy (DP-SGD), Opacus, TF-Privacy, privacy auditing and evaluation, safety classifiers, red-teaming and jailbreaking analysis
- **Tools & Systems**: Git, GitHub, Docker, Kafka, AWS, Microsoft Azure, Redis, Postman, Linux, Slurm, Jupyter, LaTeX
- **Databases**: PostgreSQL, MySQL, SQL Server MongoDB, Firebase

Awards

- **Conference of Language Modeling (COLM) Travel Grant**: 2026
- **Graduate Student Spotlight Mention for Outstanding Research Perfromance**: 2026
- **International conference on Learning Representation (ICLR) Travel Award**: 2026
- **Grace Hopper Celebration Tech Conference Travel Award**: 2025
- **Best Data Science Webinar Instructor Award**: 2023
- **Distinction in UBL Datathon for AI-based ATM Theft Detection Prototype**: 2022
- **Russian AI and Mathematics Olympiad Winner**: 2021
- **Academic High Achiever Award in 5th 6th and 8th Semester** : 2020-2021
- **National Talent Scholarship**: 2017-2021
- **Crossroad Emerging Leader Finalist by Harvard Business School**: 2020
- **PEEF Merit Based Scholarship award from Provincial Govt.**: 2015-2017
- **2nd position at District Level in Kangaroo Math Contest** : 2017
- **1st position at national level in Young Learners Math Competition** : 2017

Academic Services

- **Area Chair**: Who Verifies the Agents? Workshop @ Neurips 2026.
- **Program Committee Member**: Bidirectional human-ai-alignment Workshop @ Neurips 2026
- **Conference Reviewer**: IEEE Conference on AI (CAI) 2026; AAAI (2025, 2026), ICML 2026, COLM 2026, Neurips 2026.
- **Journal Reviewer**: Machine Intelligence Research, Transactions on Network Science and Engineering, Informatics for Health and Social Care
- **Teaching Assistant**: Introduction to Artificial Intelligence, Introduction to Algorithms, Computer Architecture and Operating Systems, Computer Organization (RPI).

Volunteer / Membership

ICLR 2026 Volunteer, Rio de Janeiro, Brazil | Microsoft Student Partner, Islamabad | UN Millennium Fellowship Campus Network Project Member | Chair, ASWAN, ACM Student Chapter